Report_DEB

1. Importation des données

```r
file_path <- here::here("mod/N_DEB_SAAWKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3311, C2 = 6666, C3 = 2212)

text_priors <- '
Distrib (Fm,           TruncNormal,           0.1,      0.2,  0.01,      0.8);
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,        0.8,      0.1,    0.1,      2);
Distrib (v,            TruncNormal_cv,    0.18505,      0.1,  0.001,    2);
Distrib (kap,          Uniform,               0,     1);
Distrib (kapA,         Uniform,               0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40,    300);
Distrib (E_coc,        TruncNormal_cv,          30,      0.2,     1,    300);
Distrib (rOM_ClxHorse, LogUniform,         0.00001,      0.3);
Distrib (K,            Uniform,            0.00001,        2);
Distrib (dv,           TruncNormal_cv,           2,      0.5,     0,     10);
Distrib (wE,           LogUniform,      0.000000001,     0.1);
Distrib (muOM,         TruncNormal_cv,       11700,      0.2,    4000,    40000);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcali DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
                 Fm.1.    pAm.1. r_pAm_pM.1.       v.1.     kap.1.    kapA.1.
Result_mode 0.46633500 130.83700  0.92382700 0.1752730 0.94720500 0.67942900
2.5%        0.34057000 102.63500  0.74269900 0.1424500 0.91753400 0.54646100
97.5%       0.59649230 163.11815  1.01055800 0.2225221 0.96827700 0.80085115
Min         0.15803100  69.68950  0.62044400 0.1335840 0.27062500 0.41118600
Max         0.70907100 171.95100  1.09661000 0.2475470 0.97279100 0.87304700
Result_mean 0.46169162 131.40015  0.87426023 0.1820790 0.94834379 0.67665692
Result_sd   0.06274439  16.21832  0.06906651 0.0215457 0.01339779 0.06586216
               Ehp.1.  E_coc.1. rOM_ClxHorse.1.       K.1.    dv.1.         wE.1.
Result_mode 100.51200 28.064300     0.004356560 0.4098060 2.208170 1.236400e-05
2.5%         78.00930 19.335790     0.002610800 0.2977851 1.428160 8.702000e-06
97.5%       152.33800 38.536500     0.007443089 0.5430940 2.788222 1.698598e-05
Min          67.21990 13.370400     0.000168070 0.2348680 1.115430 1.361450e-08
Max         168.11200 46.619100     0.011687600 0.9635530 3.466490 2.123090e-05
Result_mean 114.52755 28.793427     0.004842124 0.4255034 2.015154 1.255575e-05
Result_sd    18.47445  4.785322     0.001264830 0.0648127 0.348465 2.358138e-06
              muOM.1. Sigma_W.1.
Result_mode 11062.700 0.10619900
2.5%         5832.650 0.09599710
97.5%       14830.900 0.12262265
Min          4003.460 0.08479820
Max         19042.900 6.05999000
Result_mean 10110.909 0.10880874
Result_sd    2321.228 0.05367846

Number of observations, n =  277
Number of parameters, k =  13
Log-likelihood, LL =  0.1285206
BIC =  72.85519


Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.08       1.25
E_coc.1.              1.17       1.49
Ehp.1.                1.40       2.05
Fm.1.                 1.29       1.78
K.1.                  1.15       1.44
kap.1.                1.42       2.10
```

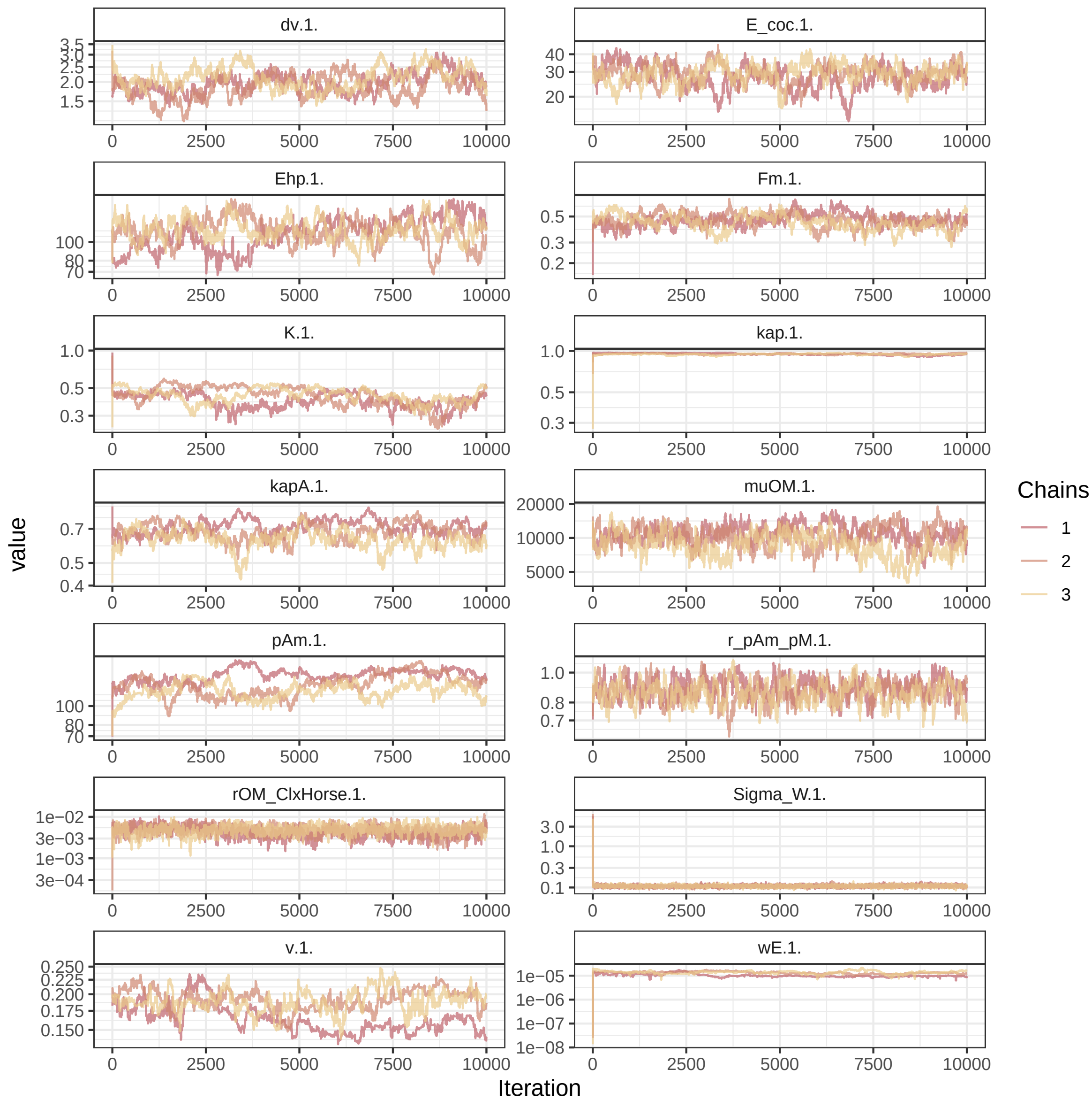


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

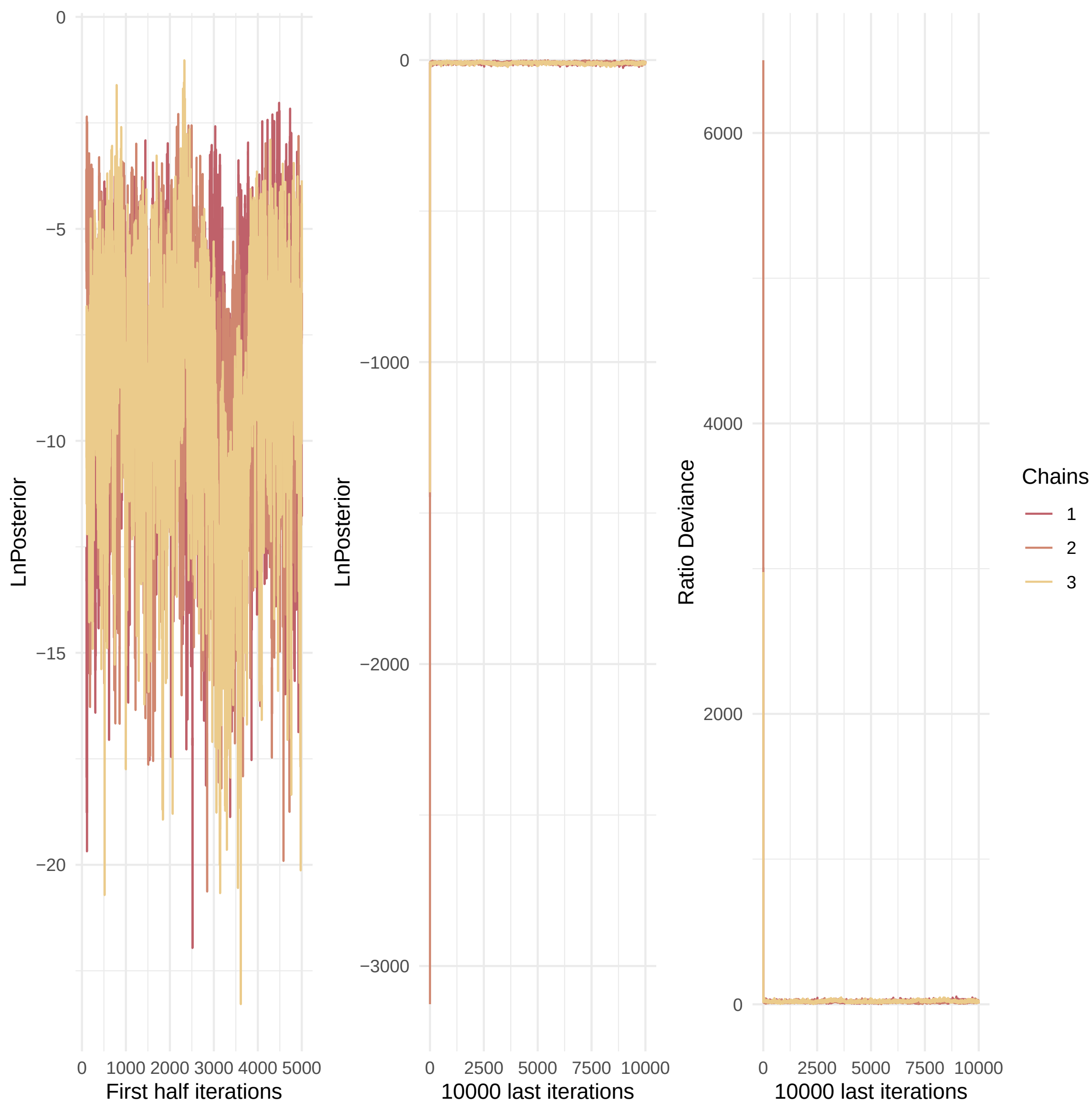


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
kapA.1.              1.73      2.77
muOM.1.              1.42      2.11
pAm.1.               2.20      4.14
r_pAm_pM.1.          1.06      1.19
rOM_ClxHorse.1.      1.21      1.62
Sigma_W.1.           1.02      1.07
v.1.                 2.60      5.23
wE.1.                2.52      8.14

Multivariate psrf

3.05
```

```r
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

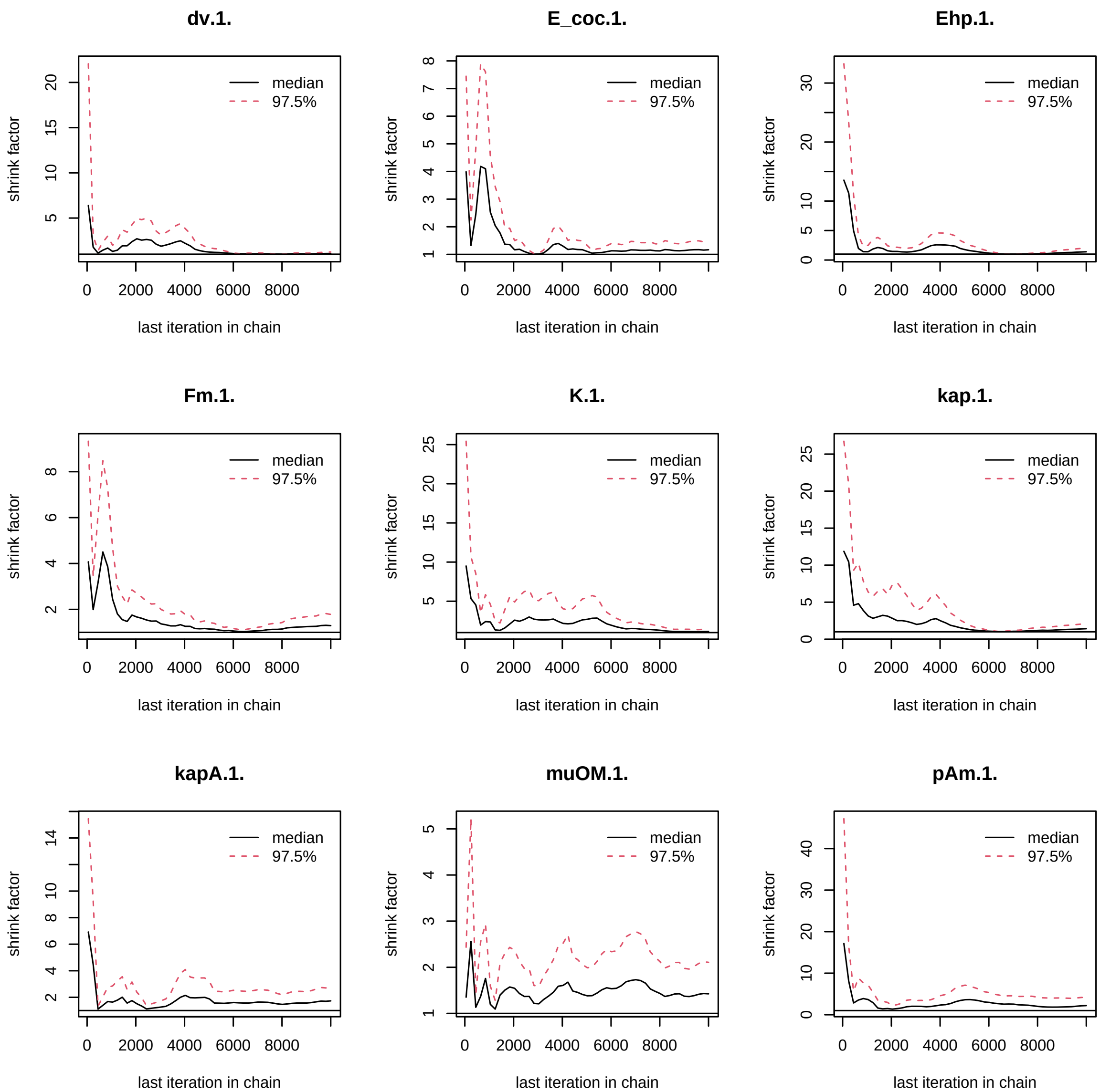


Figure 3: Chains convergence.

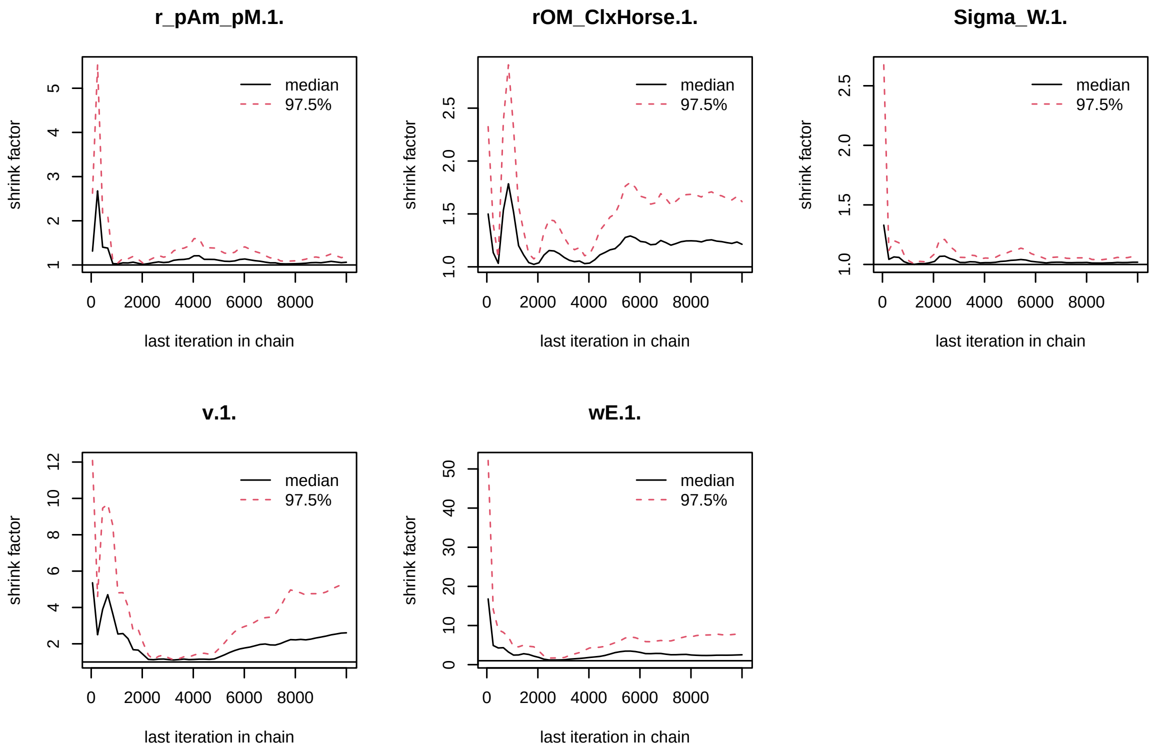


Figure 4: Chains convergence.

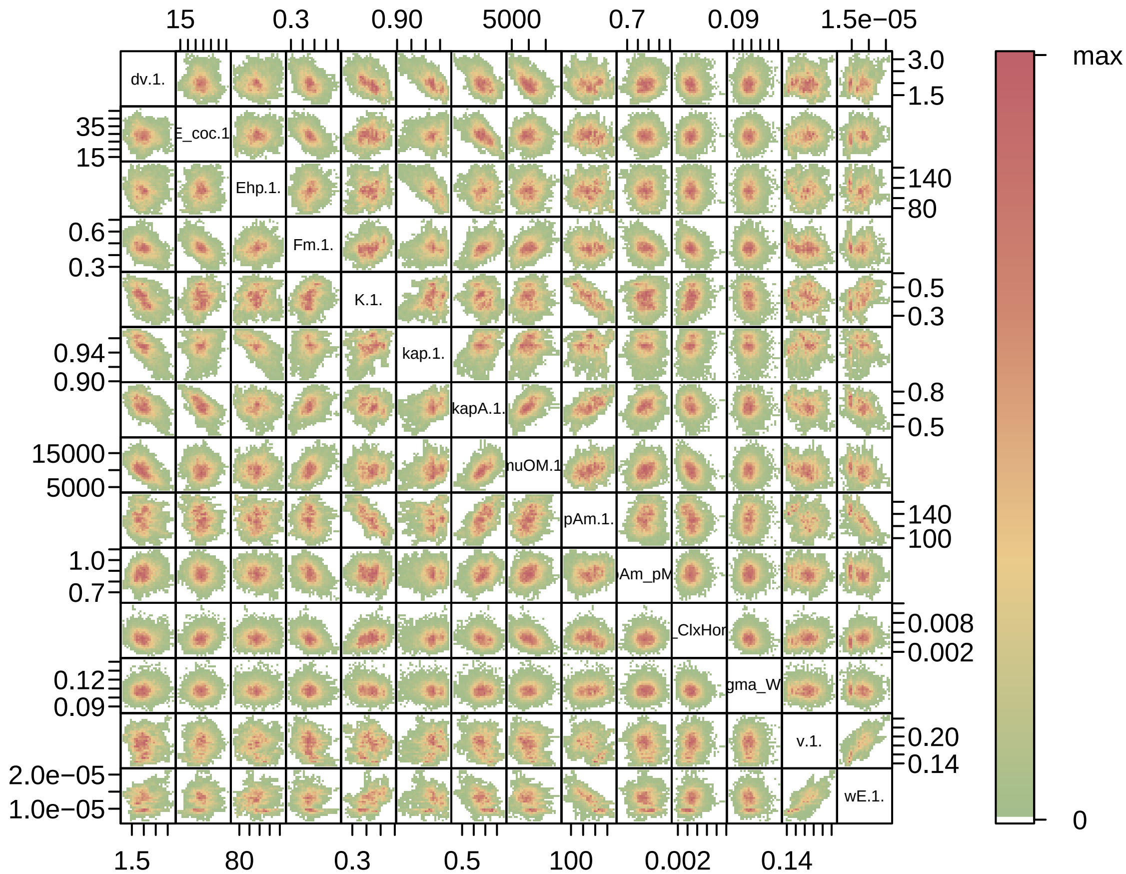


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

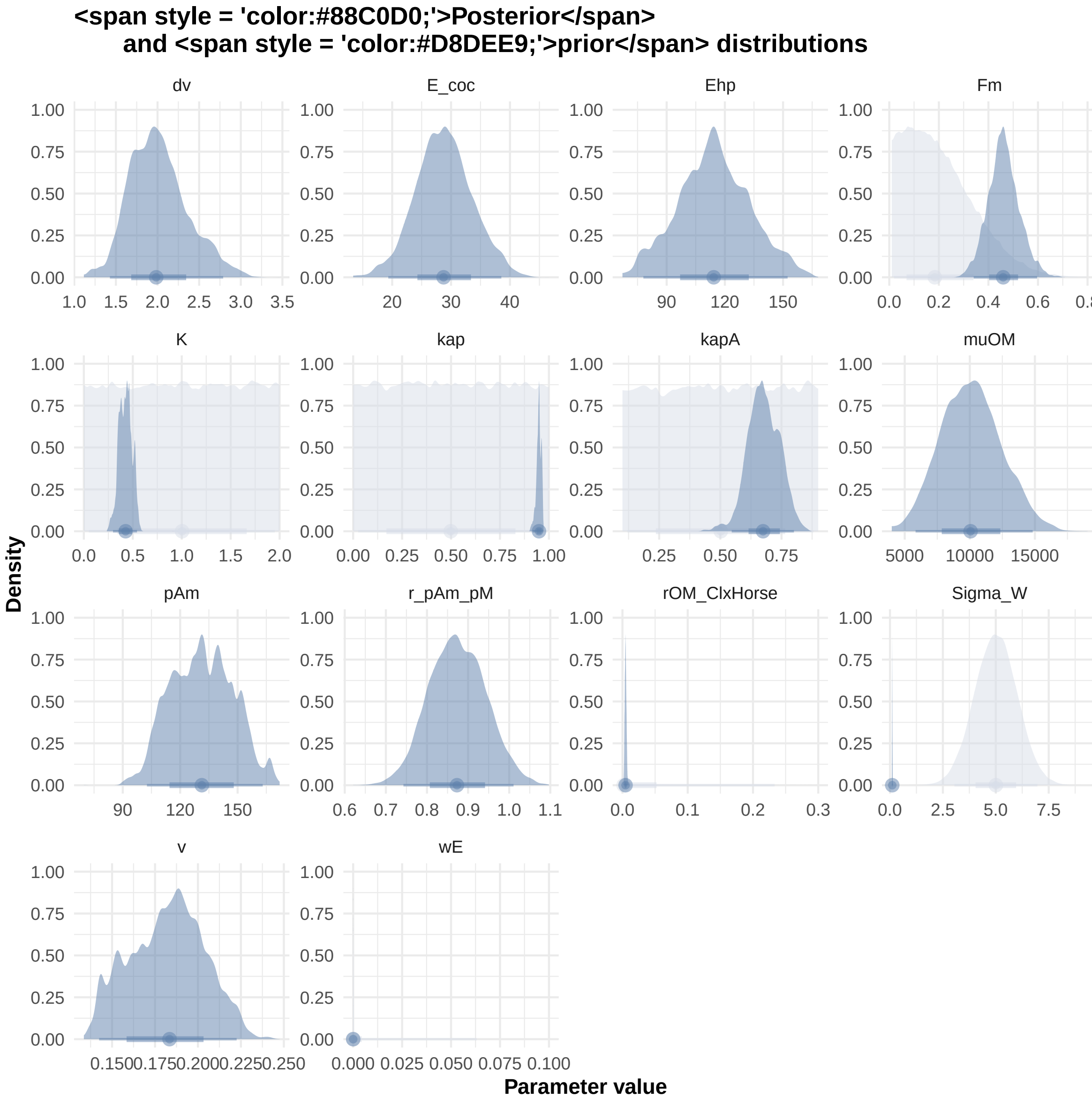


Figure 6: Distributions of the priors and the posteriors.

```
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
```

```r
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

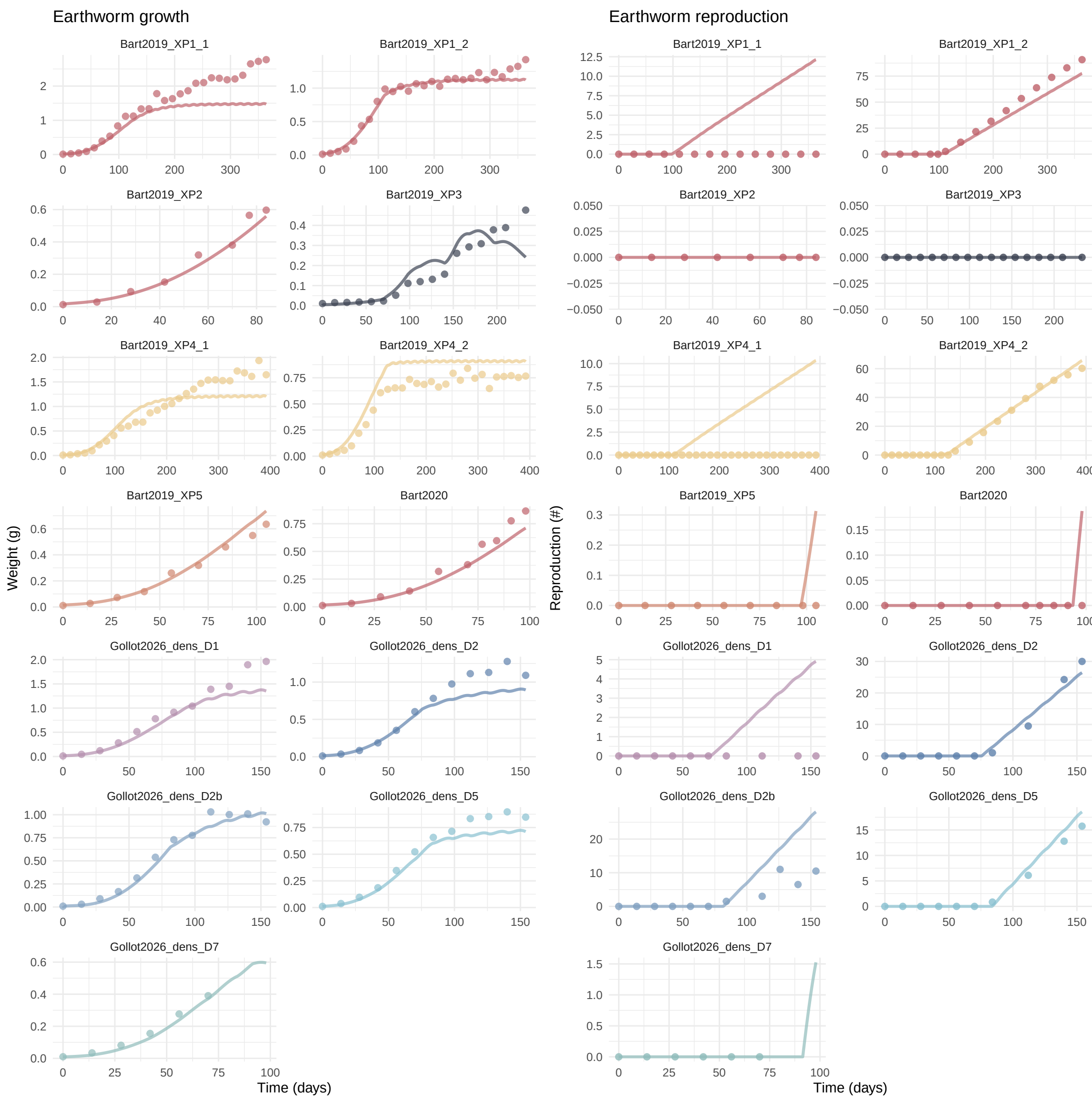
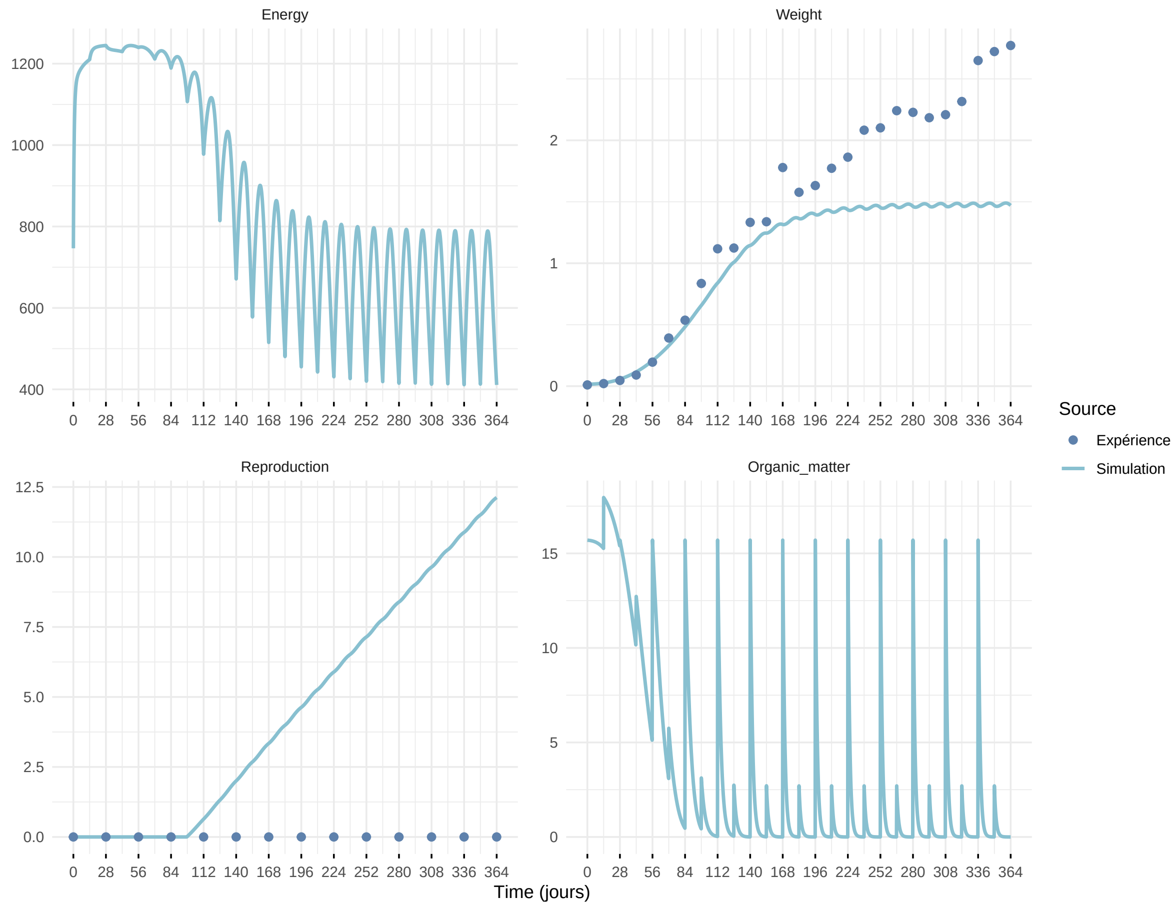
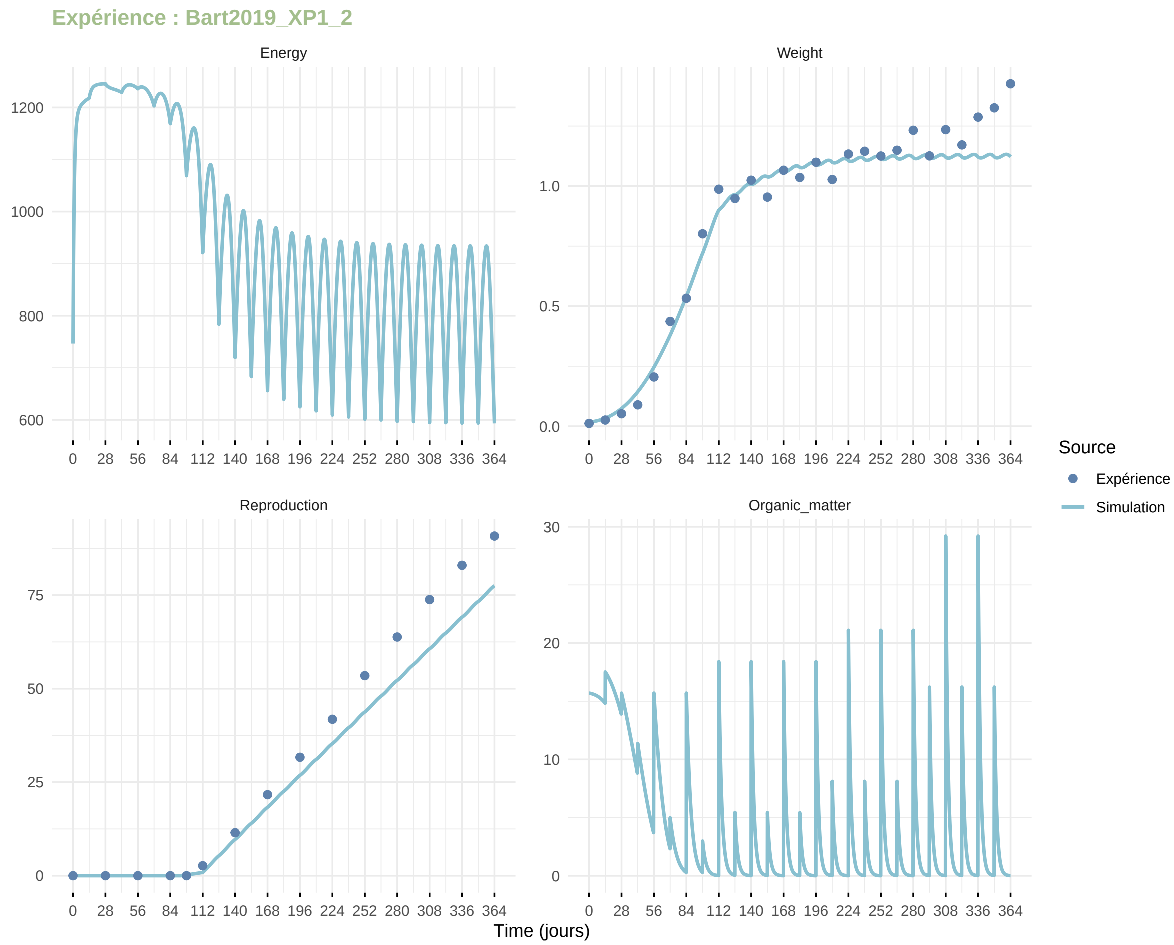


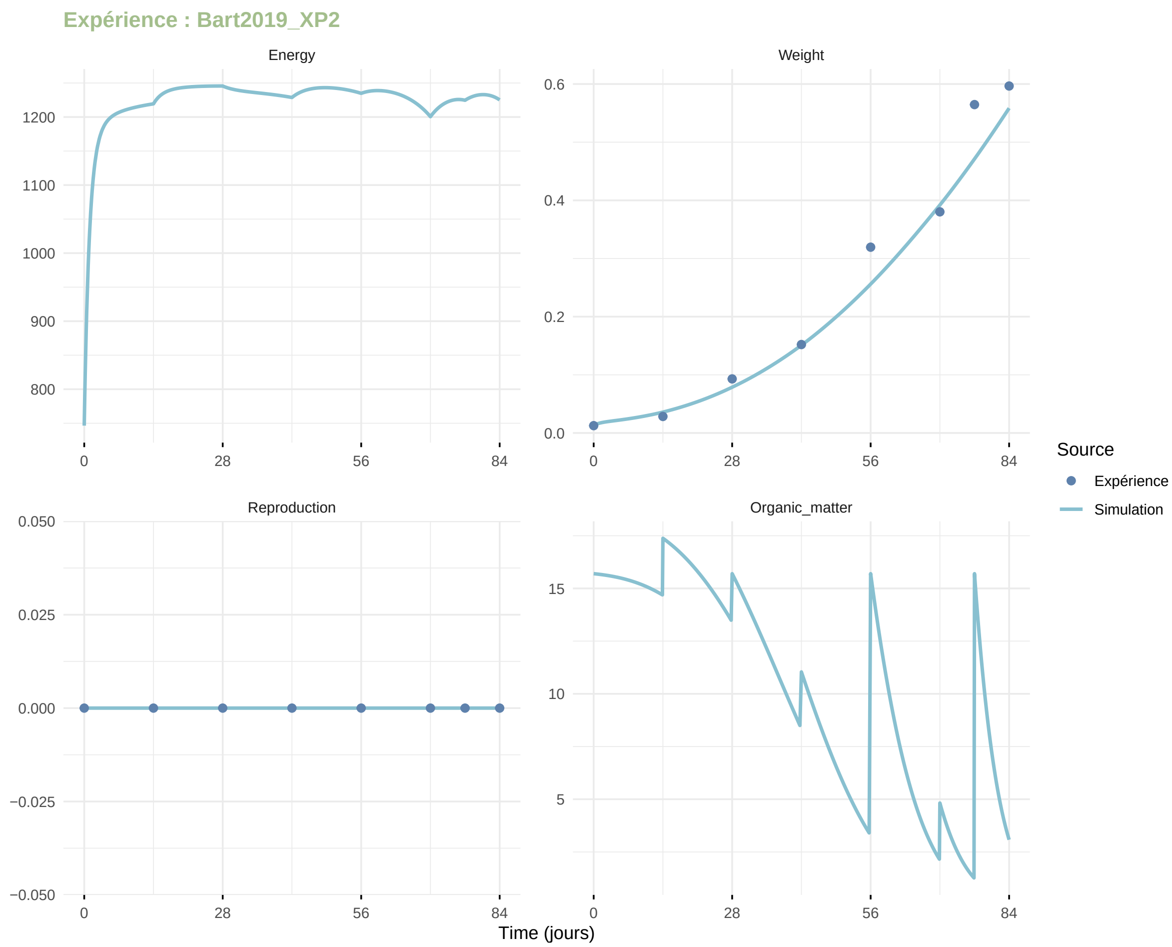
Figure 7: Simulations Weight and Reproduction.

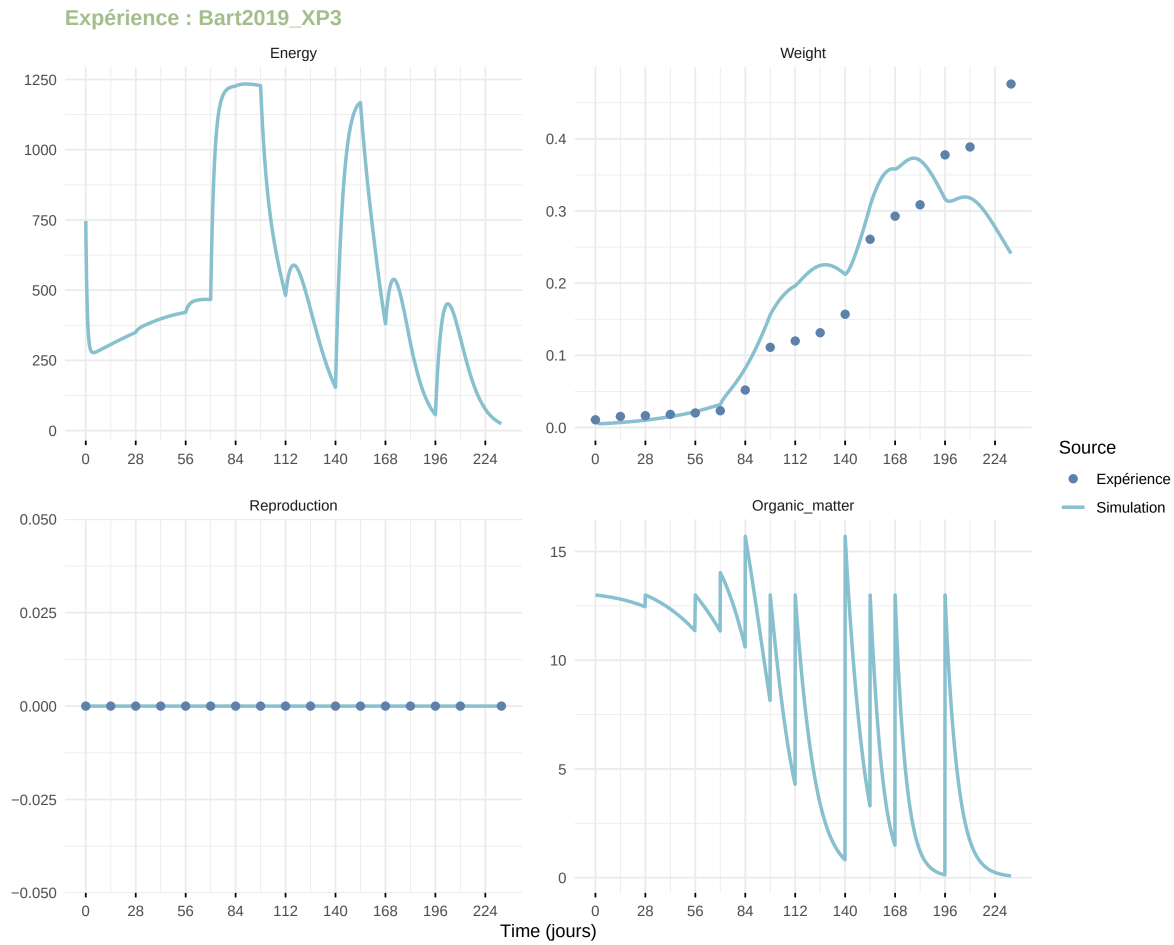
Expérience : Bart2019_XP1_1



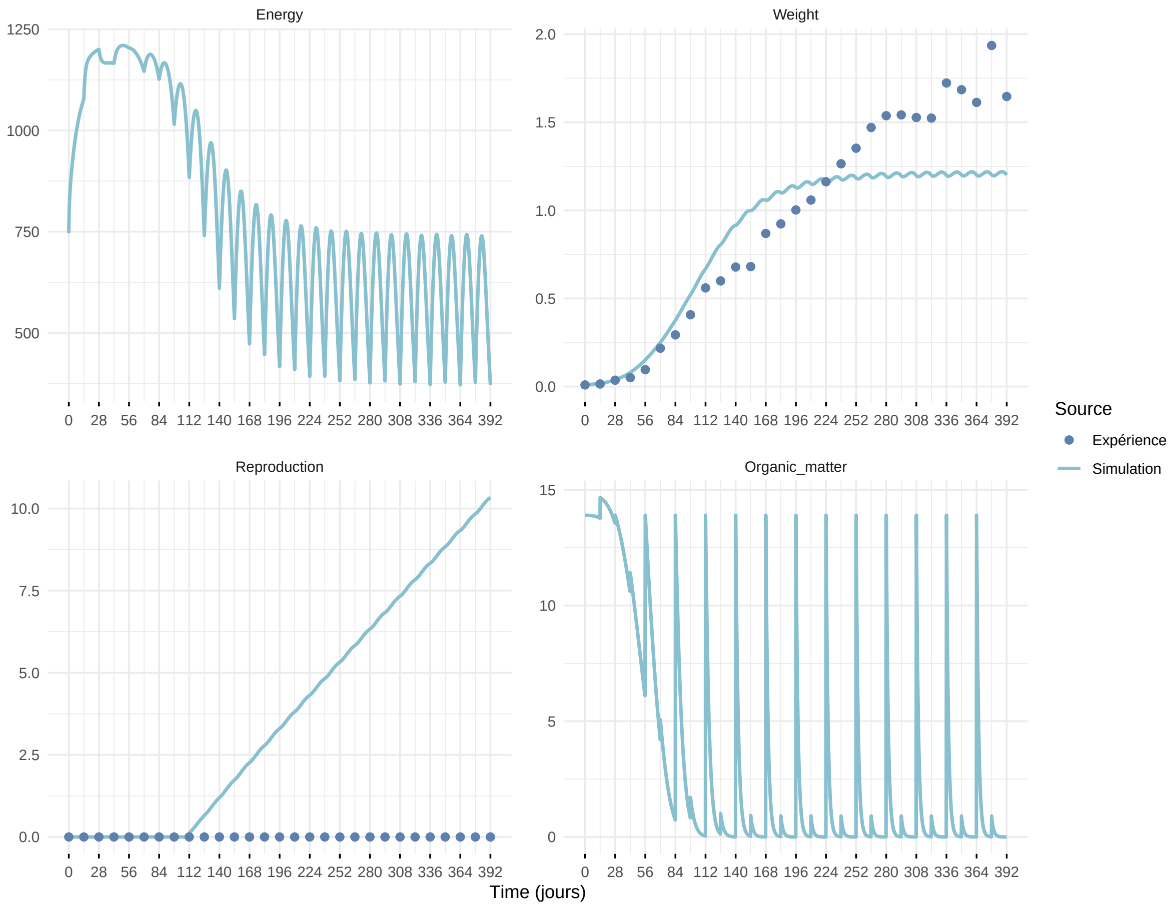


Expérience : Bart2019_XP1_2
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

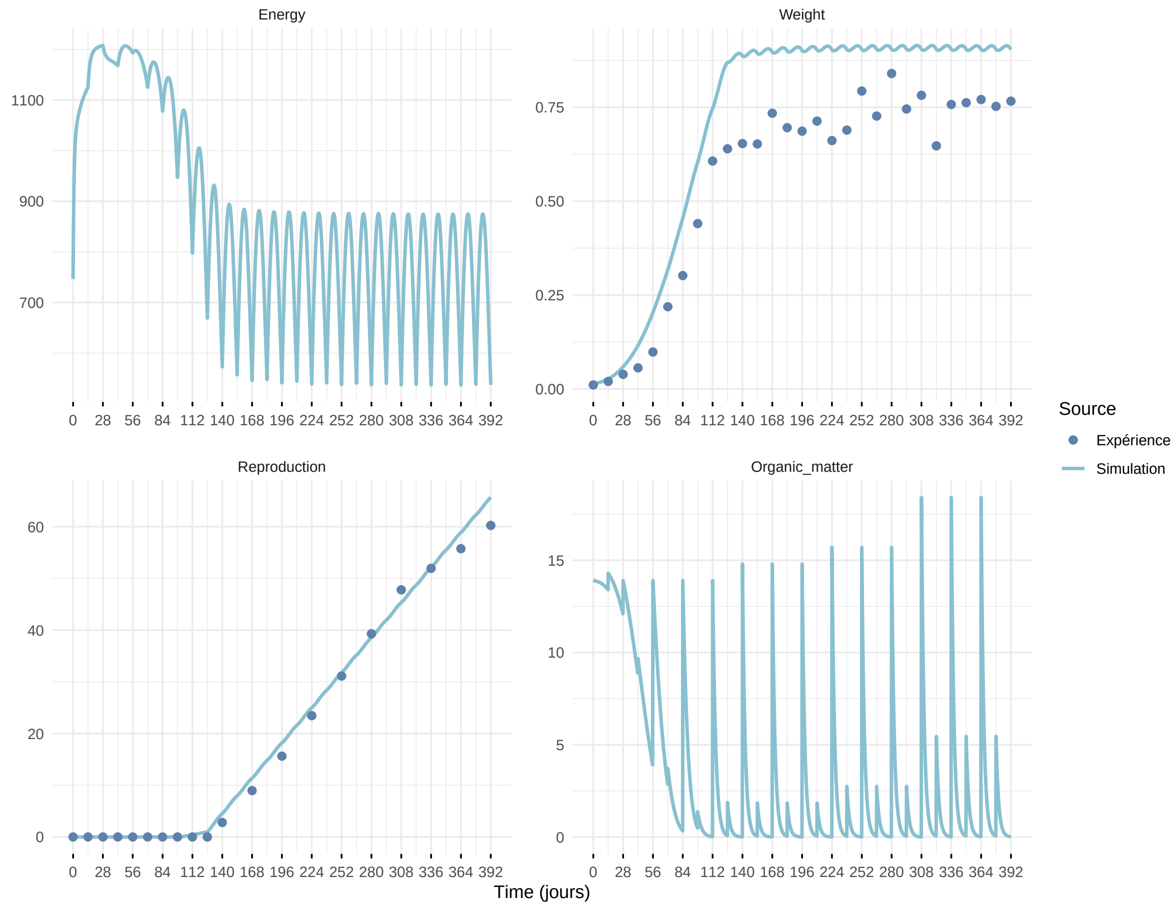

Expérience : Bart2019_XP2
Energy
1200
1100
1000
900
800
0
28
56
84
Weight
0.6
0.4
0.2
0.0
Reproduction
0.050
0.025
0.000
−0.025
−0.050
Organic_matter
15
10
5
Time (jours)
Source
Expérience
Simulation

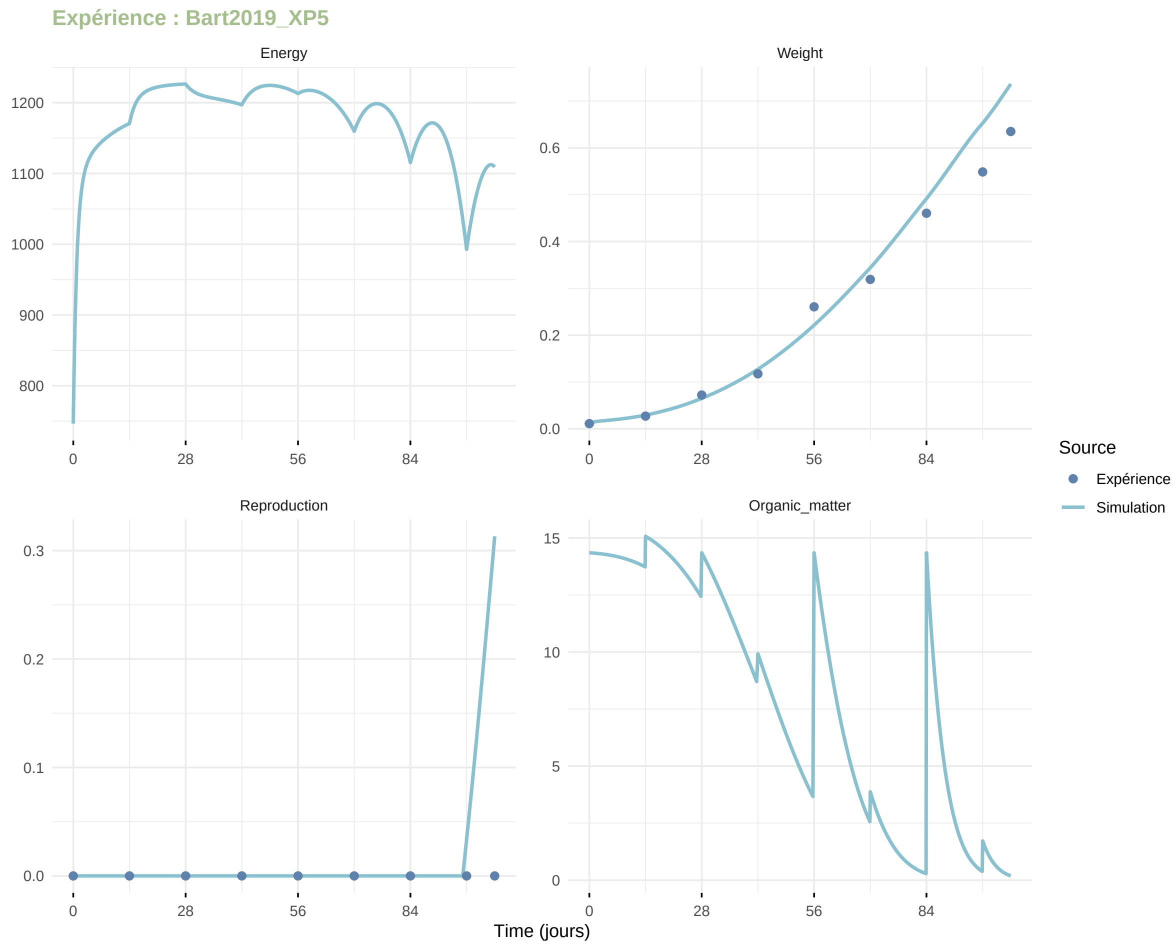

Expérience : Bart2019_XP3
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)
1250
1000
750
500
250
0
0.4
0.3
0.2
0.1
0.0
0.050
0.025
0.000
−0.025
−0.050
15
10
5
0 28 56 84 112 140 168 196 224

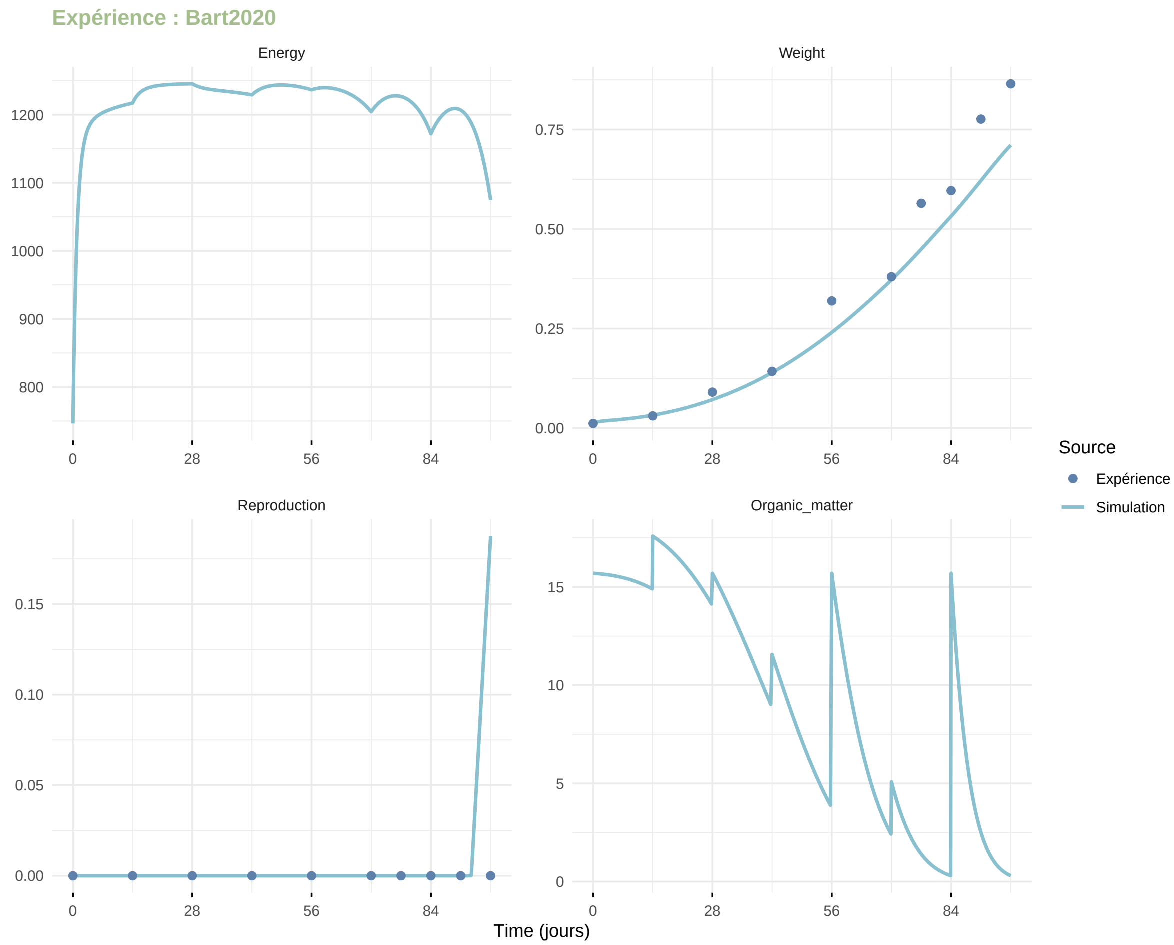
Expérience : Bart2019_XP4_1

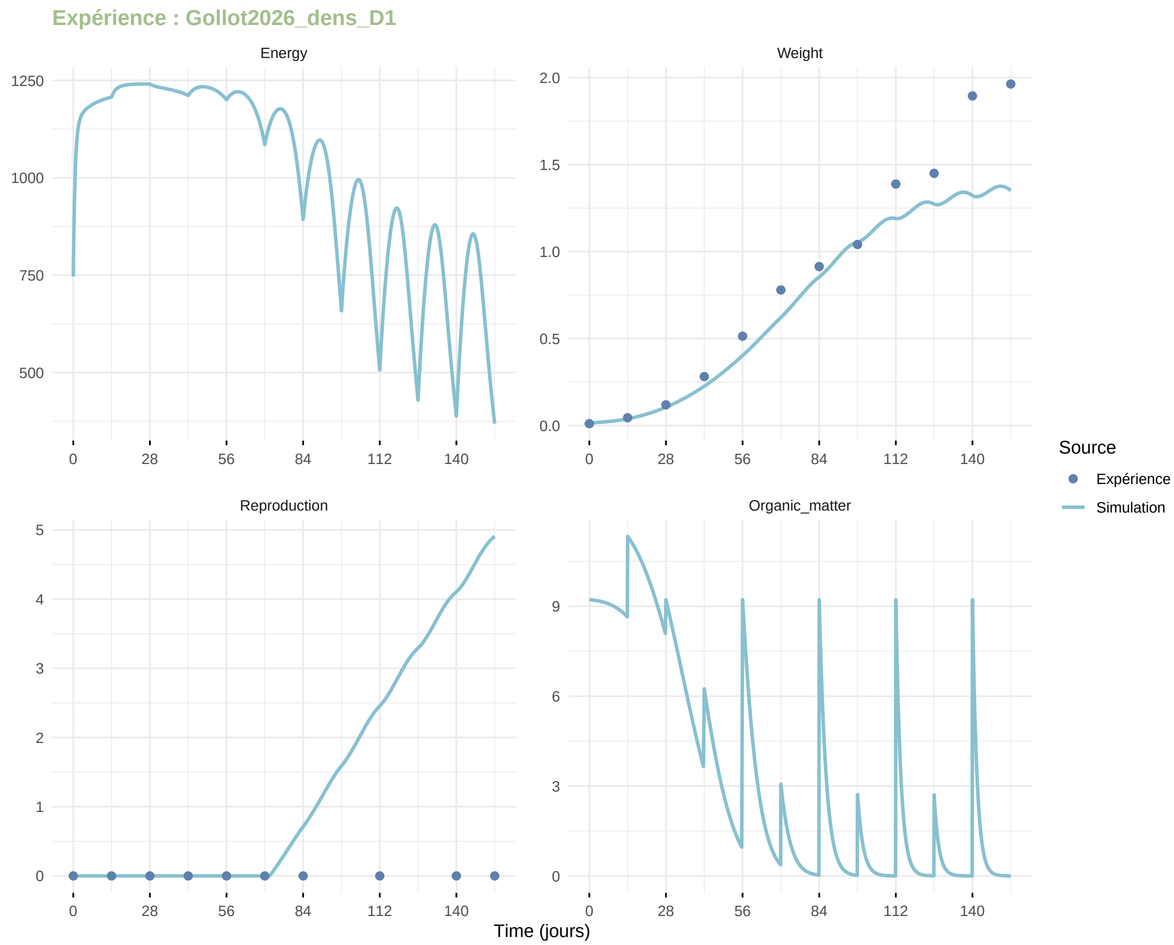


Expérience : Bart2019_XP4_2

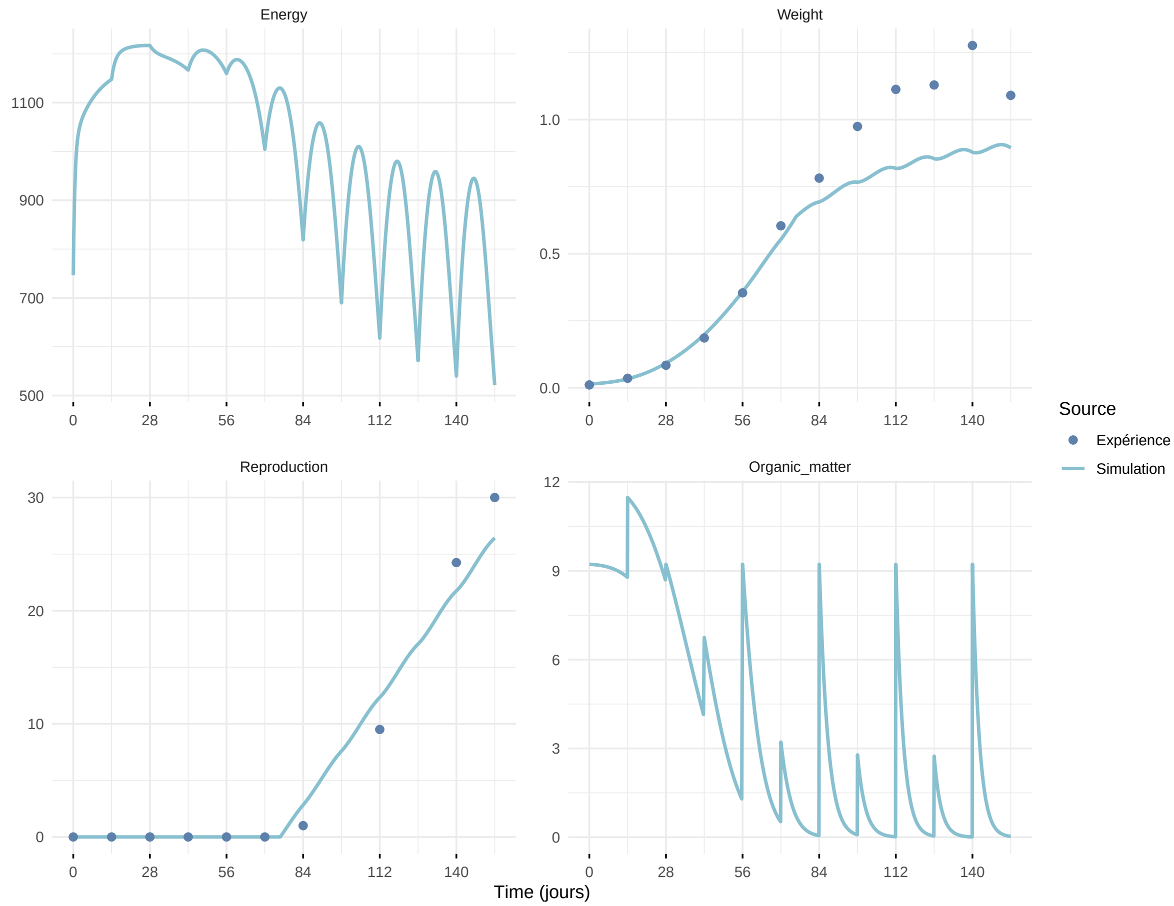



Expérience : Bart2019_XP5
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

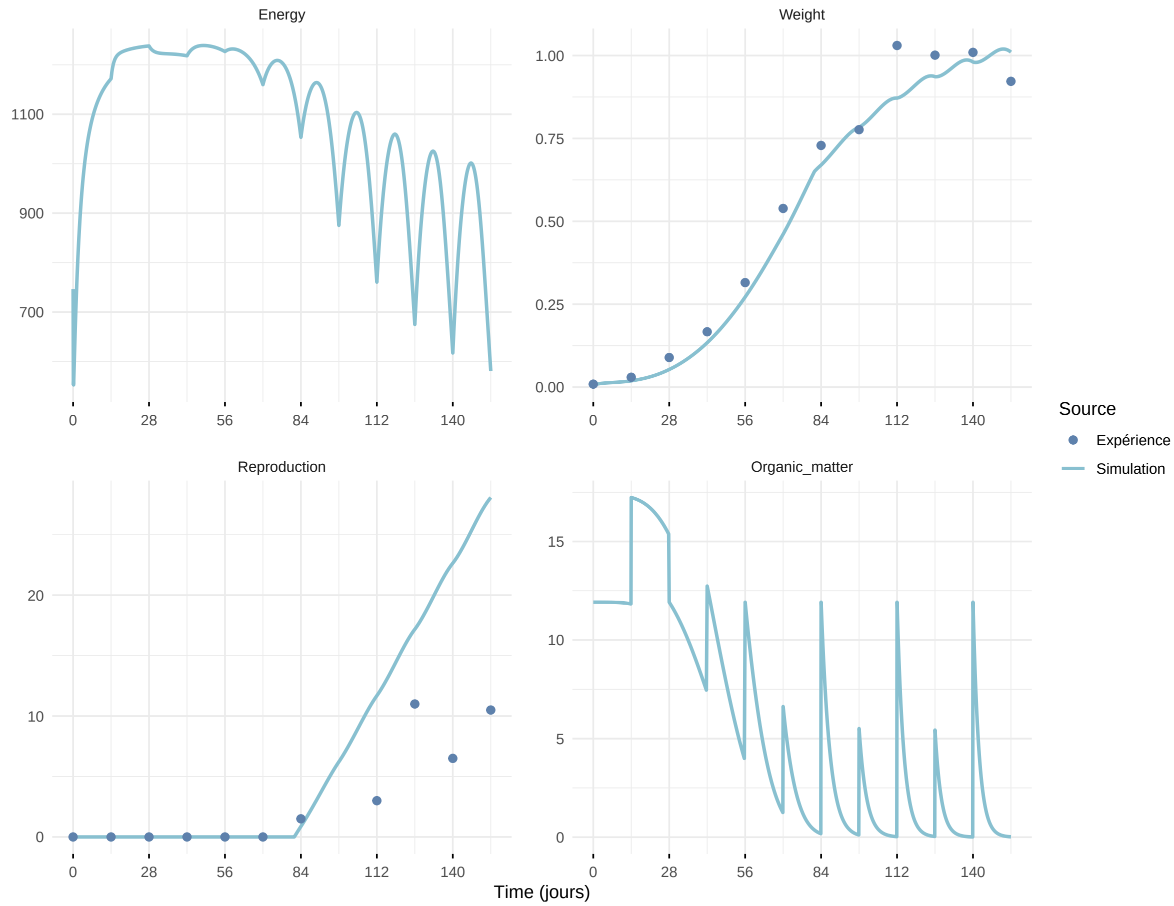

Expérience : Bart2020
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)


Expérience : Gollot2026_dens_D1
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

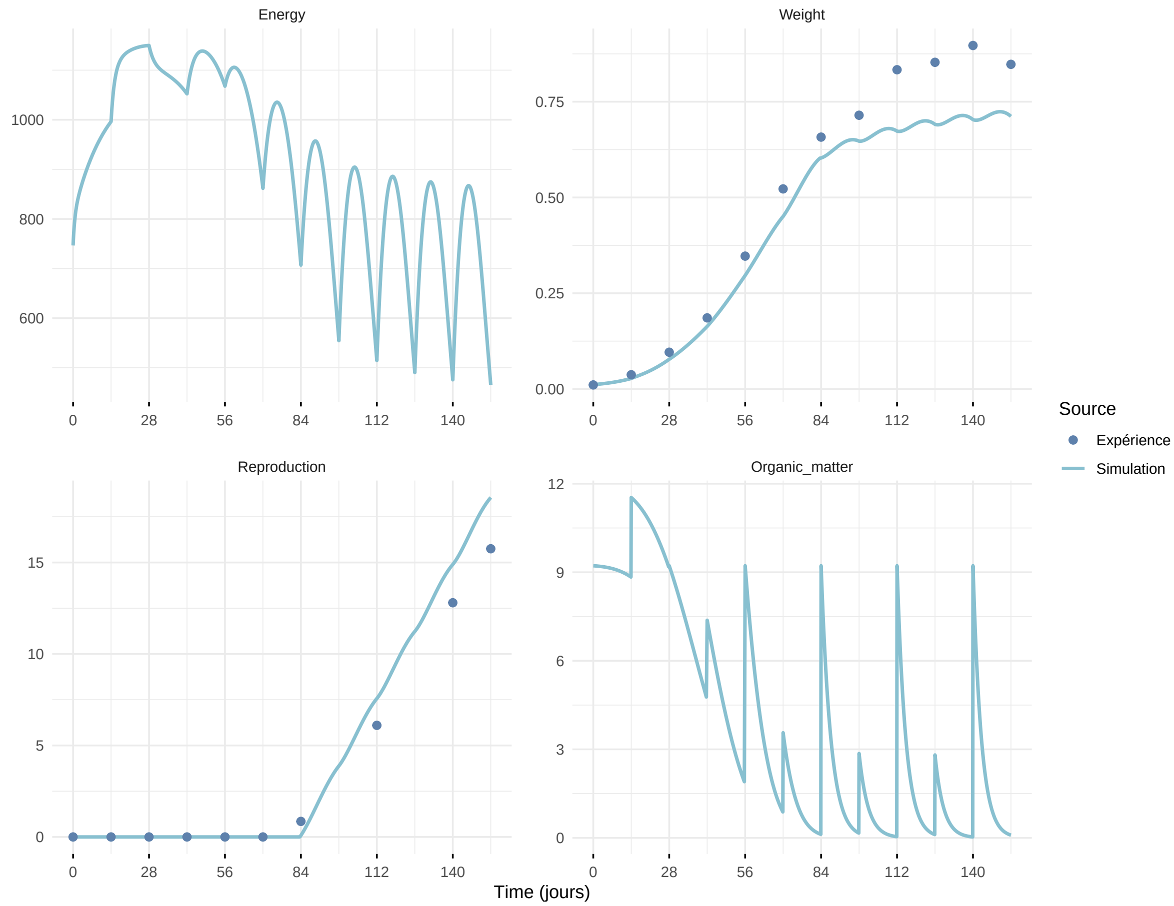
Expérience : Gollot2026_dens_D2

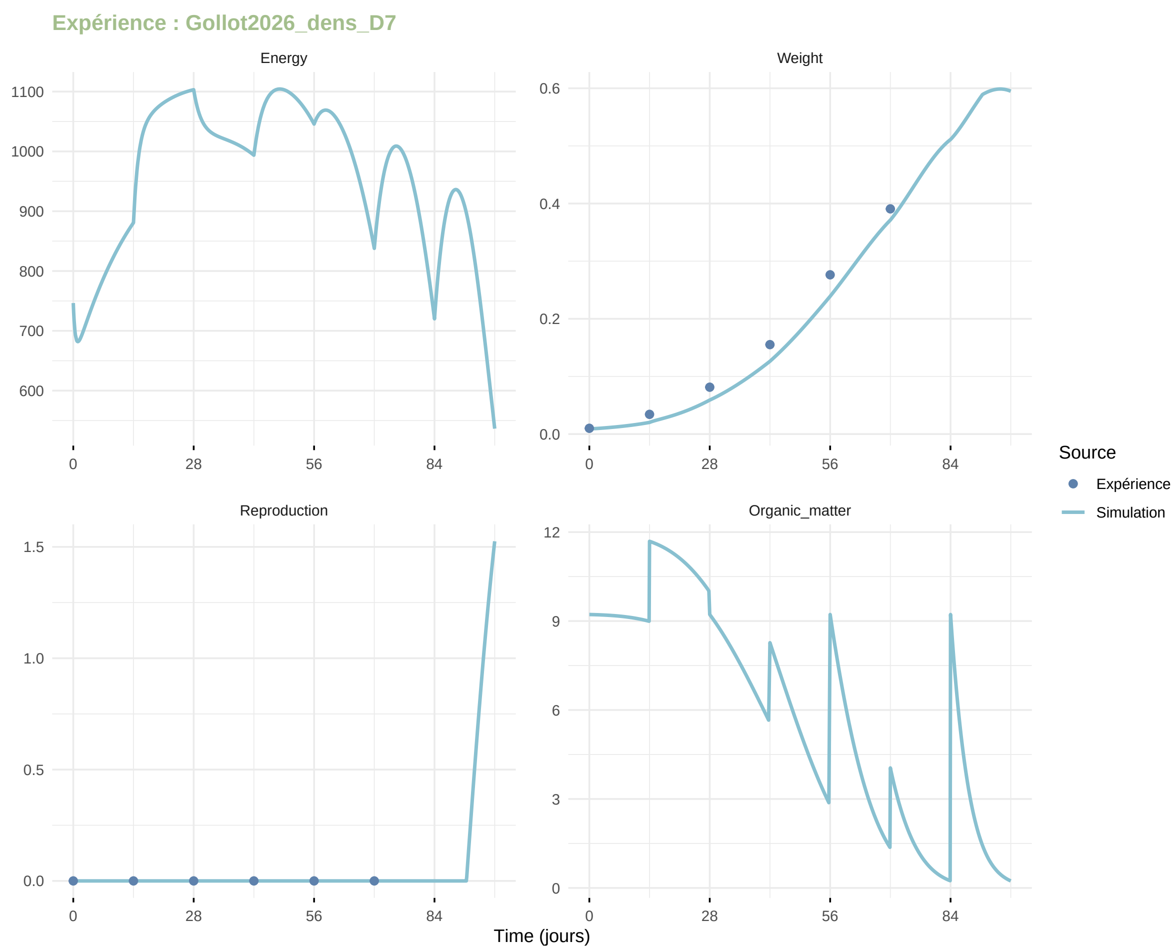


Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5




Expérience : Gollot2026_dens_D7
Energy
1100
1000
900
800
700
600
0
28
56
84
Weight
0.6
0.4
0.2
0.0
0
28
56
84
Reproduction
1.5
1.0
0.5
0.0
0
28
56
84
Organic_matter
12
9
6
3
0
0
28
56
84
Time (jours)
Source
Expérience
Simulation